

An Explorative Study of Non-Dilutive funding and Public Mega funds in Technology and Innovation Financing

The European Union's economic landscape is increasingly defined by a "productivity gap" rooted in a structural failure to scale deep-tech innovations. While the US and China have created dynamic ecosystems that facilitate high firm turnover and the rise of new industrial leaders, the EU's corporate landscape remains dominated by established incumbents.

In response, the Union has moved beyond traditional market-failure corrections, launching a series of high-stakes initiatives designed to reform its industrial policy. These include the European Innovation Council (EIC), the STEP Scale-Up instrument, and the European Chips Act. These are not mere funding programs; they are strategic attempts to foster both incremental growth in both new and mature sectors and radical breakthroughs in frontier technologies. However, this "reformist turn" faces a fundamental paradox: the EU is attempting to catalyse radical, high-uncertainty growth on a mandate for market stability, regulatory neutrality, and regional redistribution.

RQ1: *How do the innovation governance systems of the United States, China, and the European Union differ in their treatment of fundamental uncertainty versus calculable risk in the financing and regulation of deep technology development?*

RQ2: *How is the concept of "mission-oriented" innovation translated into EU administrative instruments, and to what degree does this translation emphasize regulatory and financial de-risking over technological directionality and scale?*

Learning Objectives

Program-Level Learning Objectives (MSc in Technology Entrepreneurship)

This thesis aims to demonstrate the student's ability to:

- Articulate a clearly delineated and program-relevant research topic that structures the thesis work.
- Select and apply relevant scientific theories to analyse and address the chosen topic.
- Substantiate and apply an appropriate research design and methodology.
- Ensure logical coherence between problem statement, analysis, and conclusions.
- Discuss analytical results in a critical and reflective manner.
- Extract and articulate implications for professional and policy-relevant practice.
- Communicate complex academic arguments clearly in written form.
- Argue for the significance of theoretical, methodological, and analytical choices made during the thesis process.
- Reflect on limitations of the research design, data, and analysis.
- Demonstrate the ability to work independently and reflect on academic development and specialization.

Thesis-Specific Learning Objectives

In addition, the thesis specifically aims to enable the student to:

- Frame and analyse innovation policy challenges in deep-technology domains where outcomes are shaped by fundamental uncertainty rather than calculable risk.
- Critically engage with and compare innovation governance arrangements across different political-economic systems, recognizing trade-offs, institutional constraints, and incomplete information.
- Interpret how abstract policy concepts such as “mission-oriented innovation” are translated into concrete administrative and financial instruments and assess the tensions that arise in this translation process.
- Reason analytically about how institutional priorities such as market stability, regulatory neutrality, and redistribution interact with objectives related to technological scale, speed, and industrial transformation.
- Reflect critically on the limits of comparative and institutional analysis when assessing complex, evolving innovation systems.